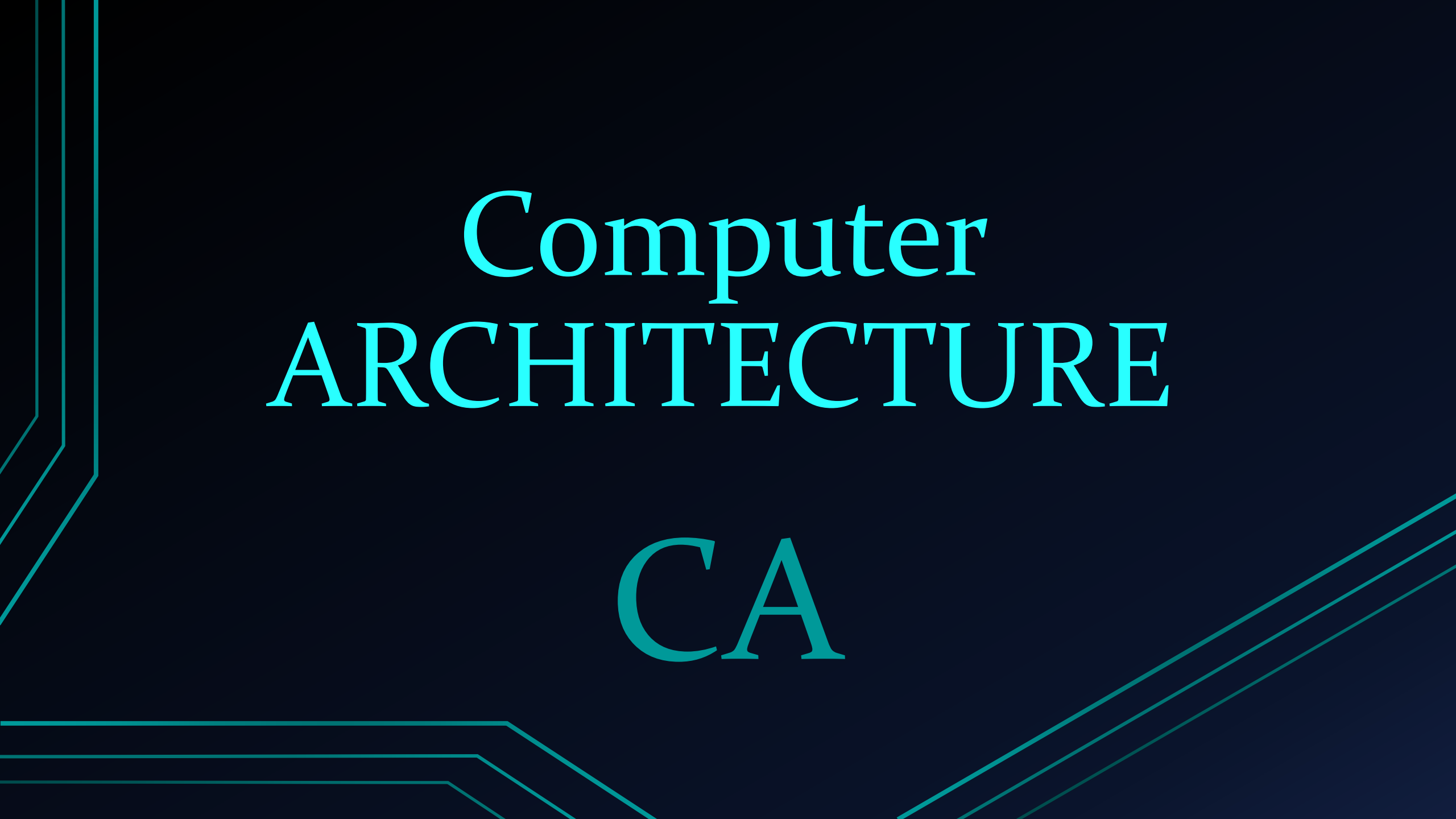




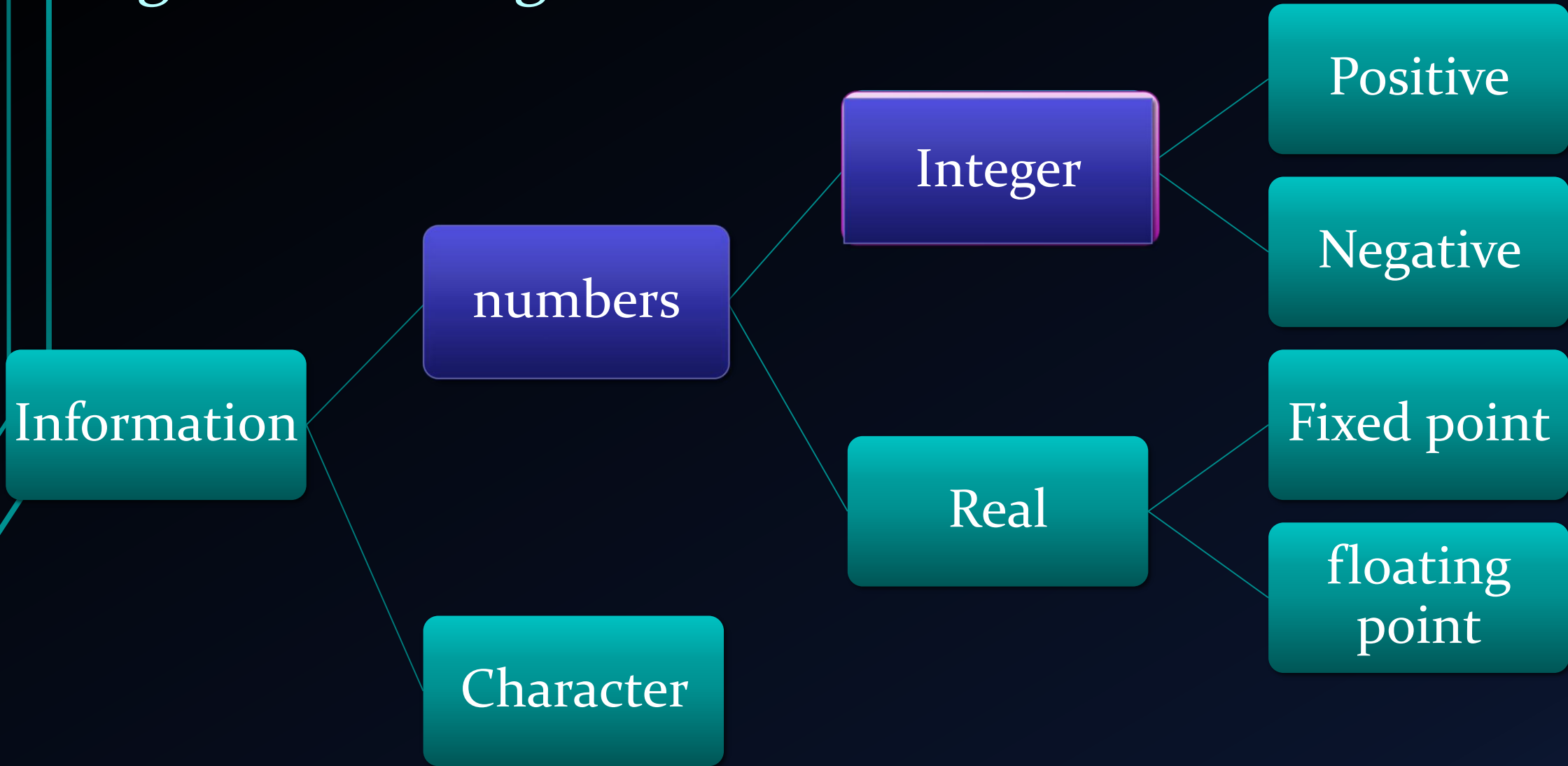
Computer ARCHITECTURE

CA

CHAPTER 2

INFORMATION REPRESENTATION

Digital encoding



Signed integers

Introduction

- An integer can be either negative or positive; we use the "-" sign to indicate a negative number and the "+" sign for a positive number.
- In binary calculations, positive integers are represented using only 0s and 1s.
- The negative sign (-) preceding integer values poses a challenge for machine representation.
- Q: How is the sign represented?
- A: We use one bit to indicate the sign."

Signed binary

- The most significant bit is used to represent the sign of the number; we call it the sign bit.
- if highest bit = 1 then negative number;
- if the highest bit = 0 then positive number;
- The other bits encode the absolute value of the number
- Example : representing decimal numbers (4) and (-4) with sign

+4	0	1	0	0
-4	1	1	0	0

Signed binary

- Note
- The use of signed numbers introduces significant restrictions.
- Before performing any operations, it is essential to define the number of bits used.
- The most significant bit is always reserved for the sign.
- For an n-bit unsigned space, the range is from 0 to $(2^n - 1)$
- There are cases where overflow can occur.

Signed binary

- Starting from the same principle, several methods are used to represent negative numbers in a computer, among which we cite:
 - Sign-magnitude;
 - 1's complement;
 - 2's complement(Used on computer) .

Sign-magnitude Method

- In the Sign-magnitude method, integers are represented in binary using $\pm$ magnitude (absolute value).
- The most significant bit (MSB) is reserved for the sign, with 1 indicating a '-' (negative) sign and 0 indicating a '+' (positive) sign.
- An integer coded on n bits will have a magnitude V :

$$0 \leq V \leq 2^{n-1}$$

Sign-magnitude Method

- Given an integer X represented in Sign-magnitude using n bits,
- the most significant bit (MSB) is reserved for the sign, while the remaining $n-1$ bits represent the magnitude.
- Since n bits allow for 2^n possible values, we have 2^{n-1} positive and 2^{n-1} negative values.
- The range for the magnitude is from 0 to $(2^{n-1}-1)$
 - The negative lowest number that can be stored is $-(2^{n-1}-1)$
 - The positive largest number that can be stored is $+(2^{n-1}-1)$.

Sign-magnitude Method

- So **zero** counts 2 times (**+0**) and (**-0**) .
- Zero then has two representations
- The range is
- $(-(2^{n-1}-1), \dots, -1, -\mathbf{0}, +\mathbf{0}, 1, 2, \dots, (2^{n-1}-1))$.

Sign-magnitude Method

- What does the following **five** bits number with signed value represent?

➤ 10000

- The following five bit number as a signed value



- What do you notice?

Sign-magnitude Method

- **Example :**
- With 4 bits how many integers can we represent?
- With 4 bits we will have **16** integers:
- $(-7, -6, -5, -4, -3, -2, -1, -0, +0, +1, +2, +3, +4, +5, +6, +7)$.
- With 5 bits how many integers can we represent?

Sign-magnitude Method

Bit Pattern	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
Unsigned	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Sign & Magnitude	+0	+1	+2	+3	+4	+5	+6	+7	-0	-1	-2	-3	-4	-5	-6	-7

Sign-magnitude Method

- This method cannot be used mathematically for 2 reasons:
 - There are 2 representations for 0.
 - Arithmetic operations cannot be performed directly. Special treatment must be made for the sign bit.

1's complement method

- Also called Logical Complement (LC), Restricted Complement or Diminished Radix Complement;
- Positive numbers are encoded the same way as in sign-magnitude.
- Negative numbers are obtained by inverting each bit of the binary code of the positive number (i.e., replacing each 0 with 1 and each 1 with 0), thus encoding its magnitude.
- The MSB is always reserved for the.

1's complement method

- What does the following 5-bits number represent in 1's C
- 11111 and 00000
- Represents -0 and +0
- -0 11111
- +0 00000

1's complement method

- Given that n bits allow for 2^n possible representations,
- For an n -bit integer in 1's Complement:
 - There are 2^{n-1} positive number
 - There are 2^{n-1} negative number.
- In the 1's Complement system, zero is counted twice, as both $+0$ and -0 .
- The range of the representation of an n -bit integer in 1's Complement
 - $-(2^{n-1} - 1) \leq X \leq +(2^{n-1} - 1)$.

1's complement method

Bit Pattern	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
Unsigned	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Sign & M	+0	+1	+2	+3	+4	+5	+6	+7	-0	-1	-2	-3	-4	-5	-6	-7
1'sC	+0	+1	+2	+3	+4	+5	+6	+7	-7	-6	-5	-4	-3	-2	-1	-0

1's complement method

➤ Limitation :

- Two different representation for 0 (+0 and -0)
- On 8 bits:
- $+0 = 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0$ and $-0 = 1\ 1\ 1\ 1\ 1\ 1\ 1\ 1$
- Multiplication and addition are less obvious.

2's complement method

- Also called True Complement or Radix Complement
- Positive numbers are encoded the same way as in sign magnitude.
- Negative numbers are represented by taking the 1's Complement of the number and then adding 1 to it, effectively obtaining the 2's Complement.
- The most significant bit is used to represent the sign (1)
- In other terms to obtain 2's C
 1. Find the 1's C of the number
 2. Add 1 to it (to the LSB)

2's complement method

- Given that n bits allow for 2^n representation, with values ranging from 0 to $2^{n-1} - 1$ providing 2^{n-1} positive values.
- The range for an n -bit integer in 2's Complement representation is
$$-(2^{n-1}) \leq X \leq +(2^{n-1} - 1).$$
- This range accounts for both positive and negative values, with $-(2^{n-1})$ as the minimum and $+(2^{n-1} - 1)$ as the maximum

Binary	Unsigned	Sign– magnitude	Ones' complement	Two's complement
0000	0	0	0	0
0001	1	1	1	1
0010	2	2	2	2
0011	3	3	3	3
0100	4	4	4	4
0101	5	5	5	5
0110	6	6	6	6
0111	7	7	7	7
1000	8	<u>–0</u>	–7	–8
1001	9	–1	–6	–7
1010	10	–2	–5	–6
1011	11	–3	–4	–5
1100	12	–4	–3	–4
1101	13	–5	–2	–3
1110	14	–6	–1	–2
1111	15	–7	0	–1

The Signed Integer: Arithmetic

Sign-magnitude Method

- Arithmetic operations are not easy.
- Example
- Perform the addition of (-1), and (2) on 4 bits

1-	1	0	0	1
2 +			1	0
-3	1	0	1	1
VA sur 4 bits		-1 + 2 = -3		

Arithmetic in 1's complement method

➤ Addition

- The carry generated by the sign bit must be added to the final result because we perform addition on complements, including the sign bit (known as carry-forward).
- Carry bit to carry forward when adding
- When adding, carry-forward includes any carry bit generated

➤ Subtraction

- Subtraction is simplified to the addition of its 1's complement

$$a - b = a + (-b).$$

Arithmetic by 2's complement

➤ Addition

➤ A carry generated by the sign bit will be **ignored**.

➤ A necessary condition to accept the result thus obtained is that a carry must also be generated by the bit just before the sign.

➤ Subtraction

➤ Subtraction is simplified to the addition of its 1's complement

$$a - b = a + (-b).$$

Arithmetic in 2's complement

- A subtraction is reduced to the addition of its 2's complement
- No carry bit carried for addition

	¹ 1	¹ 1	1	1
+			1	0
1	0	0	0	1

-1 + 2 = 1

Arithmetics

- Subtracting a number is equivalent to adding its opposite, which is represented by its 2's complement.
- To multiply two numbers, multiply their magnitudes (absolute values) and, if the numbers have different signs, take the opposite of the result.
- For division, divide the magnitudes (absolute values) and then apply the appropriate sign to the result based on the signs of the original numbers.